MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, TYPES 1N483B, 1N485B, 1N486B, 1N5194, 1N5195, 1N5196 JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon diodes. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Type	V_{RM}	V_{RWM}	I_o 1/	I_o T_A = 150°C	I_{FSM} t_p = 1/120 s	I_f	T_{op} and T_{STG}
	V(pk)	V(pk)	mA dc	mA dc	A(pk)	mA dc	°C
1N483B, 1N5194	80	70	200	50	2	650	-65
1N485B, 1N5195	200	180	200	50	2	650	to
1N486B, 1N5196	250	225	200	50	2	650	+200

1/ Derate at 1.2 mA/°C between 25°C and 150°C. Derate at 1.0 mA/°C above 150°C.

1.4 Primary electrical characteristics.

Type	V_{F1}	I_{R1} at V_{RWM}	I_{R2} at V_{RWM} T_A = 150°C
	V dc	nA dc	µA dc
1N483B, 1N5194	1.0	25	5
1N485B, 1N5195	1.0	25	5
1N486B, 1N5196	1.0	25	5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be gold plated or tinned. Where a choice of finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the diode:

a. Country of origin.

b. Manufacturer's identification.

c. Lot identification code.

d. The "1N" portion of the type designation.

(Note: Options c and d apply to types 1N5194 through 1N5196 only.)

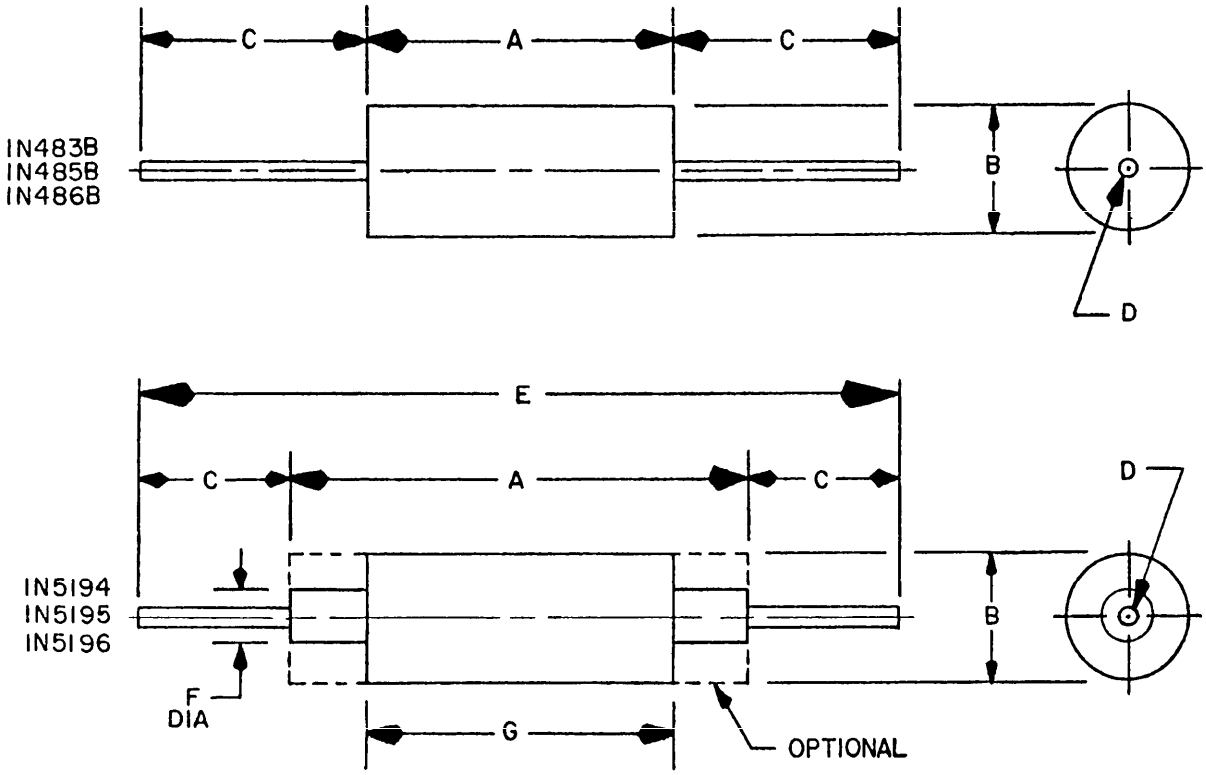
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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Type	Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 1)		Notes
		Minimum	Maximum	
1N483B	A	.230 (5.84)	.300 (7.62)	
1N485B	B	.085 (2.16)	.125 (3.18)	3
	C	1.000 (25.40)	1.500 (38.10)	4,5
1N486B	D	.018 (0.46)	.022 (0.56)	2,5
1N5194	A	.120 (3.05)	.180 (4.57)	
	B	.056 (1.42)	.094 (2.39)	
	C	1.0 (25.40)		4,5
1N5195	D	.019 (0.48)	.021 (0.53)	2,5
	E	2.120 (53.85)		
1N5196	F	.030 (0.76)		
	G	.075 (1.91)		

NOTES:
1. Metric equivalents are given for general information only.
2. The specified lead diameter applies in the zone between .050 and 1.00 inch from the diode body. Outside this zone the lead diameter is not controlled.
3. The minimum body diameter shall be maintained over .15 inch of body length.
4. Leads shall be electrically insulated from the case.
5. Both leads.

FIGURE 1. Physical dimensions.

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Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{R1} and V_{F1}	Not applicable
11	I_{R1} and V_{F1}; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = 50 mV dc change from initial reading.	I_{R1} and V_{F1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = 50 mV dc change from initial reading.	Subgroup 2 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = 50 mV dc change from initial reading.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = 25°C; I_O = 200 mA dc; f = 60 Hz

1N483B, 1N5194 V_{RWM} = 70 V(pk)
1N485B, 1N5195 V_{RWM} = 180 V(pk)
1N486B, 1N5196 V_{RWM} = 225 V(pk)

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents may specify the lead finish (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN, JANTX, JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071				---	---	---	---
Subgroup 2								
Forward voltage	4011	I_F = 100 mA(pk) (pulsed) t_p = 8.5 ms max; duty cycle ≤2%			V_{F1}	---	1.0	V(pk)
Reverse current	4016				I_{R1}			
1N483B, 1N5194		V_R = 70 V dc				---	25	nA dc
1N485B, 1N5195		V_R = 180 V dc				---	25	nA dc
1N486B, 1N5196		V_R = 225 V dc				---	25	nA dc
Reverse current	4016				I_{R2}			
1N483B, 1N5194		V_R = 80 V(pk)				---	100	µA(pk)
1N485B, 1N5195		V_R = 200 V(pk)				---	100	µA(pk)
1N486B, 1N5196		V_R = 250 V(pk)				---	100	µA(pk)
Subgroup 3								
High temperature operation:		T_A = 150°C						
Reverse current	4016				I_{R3}			
1N483B, 1N5194		V_R = 70 V dc				---	5.0	µA dc
1N485B, 1N5195		V_R = 180 V dc				---	5.0	µA dc
1N486B, 1N5196		V_R = 225 V dc				---	5.0	µA dc
Low temperature operation:		T_A = -55°C						
Forward voltage	4011	I_F = 100 mA(pk) (pulsed) t_p = 8.5 ms max; duty cycle ≤2%			V_{F2}	---	1.2	V(pk)
Subgroups 4 and 5								
Not applicable								
Subgroup 6								
Surge current	4066	I_O = 200 mA dc 10 surges at one per minute; t_p= 1/120 s; I_{FSM} = 2 A(pk)						
Electrical measurements		See table IV, steps 1 and 3						
Subgroup 7								
Not applicable								

1/ For LTPD numbers see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
a. Fine leak b. Gross leak				
Electrical measurements		See table IV, steps 1 and 3		
Decap internal visual (design verification)	2075			
Subgroup 4				
Intermittent operation life	1037	I_o = 50 mA dc; f = 60 Hz; $t_{off} = t_{on}$ = 3 minutes minimum for 2,000 cycles; T_A = 25°C		
1N483B, 1N5194 1N485B, 1N5195 1N486B, 1N5196		V_R = 70 V(pk) V_R = 180 V(pk) V_R = 225 V(pk)		
Electrical measurements		See table IV, steps 1, 3, 5, and 6		
Subgroup 5				
Accelerated steady-state operation life	1027	I_o = 200 mA dc; f = 60 Hz; T_A = 125°C		
1N483B, 1N5194 1N485B, 1N5195 1N486B, 1N5196		V_R = 70 V(pk) V_R = 180 V(pk) V_R = 225 V(pk)		
Electrical measurements		See table IV, steps 1, 3, 5, and 6		
Subgroup 6				
Not applicable				

1/ For LTPD numbers see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table IV, steps 1 and 3	
Subgroup 3			
Steady-state operation life	1027	I_O = 50 mA dc; f = 60 Hz	
1N483B, 1N5194 1N485B, 1N5195 1N486B, 1N5196		V_R = 70 V(pk) V_R = 180 V(pk) V_R = 225 V(pk)	
Electrical measurements		See table IV, steps 2 and 4	
Subgroup 4			
Decap internal visual (design verification)	2075		
Subgroup 5			
Not applicable			
Subgroup 6			
High-temperature life (non-operating)	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 2 and 4	

1/ For LTPD numbers see MIL-S-19500.

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066	See figure 1	
Subgroup 2			
Thermal shock (glass strain)	1056		
Terminal strength (lead fatigue)	2036	Test condition E	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Moisture resistance	1021		
External visual	2071		
Electrical measurements		See table IV, steps 1 and 3	
Subgroup 3			
Shock	2016		
Vibration, variable frequency (monitored)	2057		
1N483B, 1N5194 1N485B, 1N5195 1N486B, 1N5196		V_R = 40 V dc V_R = 100 V dc V_R = 125 V dc	
Monitor during test: Change in reverse current of $\geq$10 µs		Maximum ΔI_R shall be 10 µA dc	
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1 and 3	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			
Steady-state operation life	1026	I_O = 50 mA dc; f = 60 Hz	
1N483B, 1N5194 1N485B, 1N5195 1N486B, 1N5196		V_R = 70 V(pk) V_R = 180 V(pk) V_R = 225 V(pk)	
Electrical measurements		See table IV, steps 1, 3, 5, and 6 (JANS) and steps 2 and 4 (JAN, JANTX, and JANTXV)	

1/ For LTPD numbers see MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1	Forward voltage	4011	I_F = 100 mA(pk) (pulsed); t_p = 8.3 ms max; duty cycle ≤2%	V_{F1}	---	1.0	V(pk)
2	Forward voltage	4011	I_F = 100 mA(pk) (pulsed); t_p = 8.3 ms max; duty cycle ≤2%	V_{F1}	---	1.2	V(pk)
3	Reverse current	4016		I_{R1}			
	1N483B, 1N5194		V_R = 70 V dc		---	25	nA dc
	1N485B, 1N5195		V_R = 180 V dc		---	25	nA dc
	1N486B, 1N5196		V_R = 225 V dc		---	25	nA dc
4	Reverse current	4016		I_{R1}			
	1N483B, 1N5194		V_R = 70 V dc		---	50	nA dc
	1N485B, 1N5195		V_R = 180 V dc		---	50	nA dc
	1N486B, 1N5196		V_R = 225 V dc		---	50	nA dc
5	Forward voltage	4011	I_F = 100 mA(pk) (pulsed); t_p = 8.3 ms max; duty cycle ≤2%	ΔV_{F1}	±50 mV dc change from initial reading		
6	Reverse current	4016		ΔI_{R1}	100% or 10 nA dc change from initial reading, whichever is greater		
	1N483B, 1N5194		V_R = 70 V dc				
	1N485B, 1N5195		V_R = 180 V dc				
	1N486B, 1N5196		V_R = 225 V dc				

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